ELEMENTARY POSITIVITY OF WEIGHTED CHROMATIC FUNCTIONS AND DUAL DYCK SYMMETRIC FUNCTIONS

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Abstract. We prove coefficientwise elementary positivity for the dual step-τ Dyck factorization series of every finite multiset of integers and every integer $\tau \geq 0$. The corresponding affine series are coefficientwise positive in the complete homogeneous basis. With an explicit change of statistic, the step-one specialization proves elementary positivity of Hawkes's dual Dyck symmetric functions and homogeneous positivity of his affine Dyck symmetric functions. The main algebraic result is a weighted extension of chromatic elementary positivity: for a natural unit interval graph, a vertex of weight m receives a set of m colors, and the resulting inversion-refined chromatic function is elementary-positive over $\mathbb{Z}_{\geq 0}[u]$. We use OpenAI's positive common-history theorem as an external input. The extension replaces a squarefree vertex target by an arbitrary multiplicity target and proves the associated coefficient identity directly. A squarefree color coefficient then forces each positive Laurent history weight to have finite support in the correct grading. This avoids division by quantum factorials. We also obtain palindromicity, a history formula for the elementary coefficients, and positivity for generalized Hessenberg chromatic functions.

1. Introduction and main results

The dual and affine Dyck symmetric functions introduced by Hawkes [**?**] arise from factorizations of a multiset into integer sequences satisfying complementary local inequalities. His insertion correspondence proves Schur positivity of the dual functions; an involution gives the conjugate-shape expansion for the affine functions [**?**, Theorem 3.18 and Corollaries 3.19–3.20]. The present paper strengthens these conclusions to positivity in the elementary and complete homogeneous bases, respectively. We work throughout with an integer step $\tau \geq 0$, obtaining the original functions as a step-one corollary with their original statistic.

A dual factor has strictly increasing entries whose successive gaps exceed τ. Its set of values is therefore an independent set in the graph joining distinct values at distance at most τ. When a value occurs repeatedly in the multiset, its occurrences lie in distinct factors. Thus the appropriate graph model is a *multicoloring*: a vertex of weight m receives a set of m colors. This weighted chromatic construction is already present in Hwang [**?**] and Kiem–Lee [**?**, Definitions 4.1–4.3]. Numbered references to Hawkes and Kiem–Lee use the respective arXiv versions listed in the bibliography. It is not the alternative vertex-weighted construction that assigns one color, raised to a vertex-dependent power, to each vertex.

The unweighted chromatic quasisymmetric functions of natural unit interval graphs were introduced by Shareshian and Wachs [**?**]. Our proof uses the common-history machinery underlying OpenAI's elementary-positivity argument [**?**, **?**]. The intermediate theorem needed here is more flexible than the final unweighted statement: it transports a general root-cone coefficient to a positive combination of ordinary elementary coefficients. We exploit this flexibility at a target with arbitrary vertex multiplicities.

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1.1. **The weighted theorem.** A *natural unit interval graph* on $[\ell] = \{1, \dots, \ell\}$ is a graph specified by a weakly increasing function $h : [\ell] \to [\ell]$ with $h(i) \geq i$, where, for $i < j$,

$$ij \in E(G) \quad \Longleftrightarrow \quad j \leq h(i).$$

Fix positive integer weights $\mathbf{m} = (m_1, \dots, m_\ell)$ and write

$$N = \sum_i m_i, \qquad A = A(G, \mathbf{m}) = \sum_{\substack{i<j \\ ij \in E(G)}} m_i m_j. \tag{1.1}$$

A proper set coloring γ assigns a set $\gamma_i \subset \mathbb{Z}_{>0}$ of cardinality m_i to vertex i, with $\gamma_i \cap \gamma_j = \varnothing$ whenever $ij \in E(G)$. Put

$$\mathrm{inv}_G(\gamma) = \sum_{\substack{i<j \\ ij \in E(G)}} \#\{(a, b) \in \gamma_i \times \gamma_j : a > b\}, \tag{1.2}$$

$$C^-_{G,\mathbf{m}}(\mathbf{x}; u) = \sum_{\gamma \text{ proper}} u^{\mathrm{inv}_G(\gamma)} \prod_i \prod_{a \in \gamma_i} x_a. \tag{1.3}$$

The ascent version $C^+_{G,\mathbf{m}}$ uses $a < b$ in (**??**). These are homogeneous quasisymmetric series of degree N with polynomial coefficients in u.

Theorem A (Weighted chromatic elementary positivity)**.** *For every natural unit interval graph G and every positive integer weight vector $\mathbf{m}$, there are polynomials $B_{G,\mathbf{m},\lambda}(u) \in \mathbb{Z}_{\geq 0}[u]$, indexed by $\lambda \vdash N$, such that*

$$C^+_{G,\mathbf{m}}(\mathbf{x}; u) = C^-_{G,\mathbf{m}}(\mathbf{x}; u) = \sum_{\lambda \vdash N} B_{G,\mathbf{m},\lambda}(u) e_\lambda(\mathbf{x}). \tag{1.4}$$

Every coefficient polynomial has degree at most A and satisfies

$$B_{G,\mathbf{m},\lambda}(u) = u^A B_{G,\mathbf{m},\lambda}(u^{-1}). \tag{1.5}$$

In particular, symmetry is part of the conclusion, not an input to our proof. For $\mathbf{m} = (1, \dots, 1)$, this recovers the unweighted result. The general weights will be essential for multisets with repeated values.

1.2. **The Dyck theorem.** Write a finite multiset of integers as

$$S = c_1^{m_1} \cdots c_\ell^{m_\ell}, \qquad c_1 < \cdots < c_\ell, \qquad |S| = N.$$

For a fixed integer $\tau \geq 0$, define

$$A_\tau(S) = \sum_{\substack{i<j \\ c_j - c_i \leq \tau}} m_i m_j, \tag{1.6}$$

$$\beta_\tau(S) = \tau \sum_i \binom{m_i}{2} + \sum_{\substack{i<j \\ c_j - c_i \leq \tau}} (\tau - c_j + c_i) m_i m_j. \tag{1.7}$$

The dual and affine factorization series defined in Section **??** are denoted by $\mathcal{D}^*_{S,\tau}(\mathbf{x}; u)$ and $\mathcal{D}_{S,\tau}(\mathbf{x}; u)$. Their coefficient of u^d counts factorizations whose concatenation has statistic $\mathrm{dinv}_\tau = d$.

Theorem B (Step-τ Dyck positivity)**.** *For every finite multiset S of integers and every integer $\tau \geq 0$, there are nonnegative integers $a_{\lambda,k}(S,\tau)$ such that*

$$\mathcal{D}^*_{S,\tau}(\mathbf{x};u) = \sum_{\lambda \vdash N} \sum_{k=0}^{A_\tau(S)} a_{\lambda,k}(S,\tau) u^{\beta_\tau(S)+k} e_\lambda(\mathbf{x}), \tag{1.8}$$

$$\mathcal{D}_{S,\tau}(\mathbf{x};u) = \sum_{\lambda \vdash N} \sum_{k=0}^{A_\tau(S)} a_{\lambda,k}(S,\tau) u^{\beta_\tau(S)+k} h_\lambda(\mathbf{x}). \tag{1.9}$$

Moreover,

$$a_{\lambda,k}(S,\tau) = a_{\lambda,A_\tau(S)-k}(S,\tau). \tag{1.10}$$

Thus each fixed-statistic dual function is e-positive and each fixed-statistic affine function is h-positive. Both are zero outside $\beta_\tau(S) \leq d \leq \beta_\tau(S) + A_\tau(S)$.

For $S = \varnothing$, the assertion means that both series are 1, with $a_{\varnothing,0} = 1$. The nonempty case occupies the proof. At step one, the factor conditions agree with Hawkes's definitions, but our statistic includes the constant $\kappa(S) = \sum_i \binom{m_i}{2}$. After this shift is removed, Theorem **??** gives the following precise classical consequence.

Corollary 1.1. *For every finite multiset S of integers and every $d \geq 0$, the functions* $\mathrm{DS}^*(S,d;\mathbf{x})$ *and* $\mathrm{DS}(S,d;\mathbf{x})$ *of* [**?**, Definition 2.5] *are respectively e-positive and h-positive, with nonnegative integer coefficients.*

1.3. **Why the multiplicity extension is needed.** Replace each vertex i of G by an ordered clique of m_i vertices, and replace each original edge by all edges between the corresponding cliques. For the resulting graph $\widetilde{G}$, the standard blowup identity is

$$X_{\widetilde{G}}(\mathbf{x};u) = \left(\prod_i [m_i]_u!\right) C^+_{G,\mathbf{m}}(\mathbf{x};u), \qquad [m]_u! = \prod_{j=1}^{m} (1 + u + \cdots + u^{j-1}). \tag{1.11}$$

See [**?**] and [**?**, Remark 4.4 and (4.3)]; a direct proof is included below. Elementary positivity of the left side does not imply coefficientwise positivity after division by the factor on the right. Even

$$(1+u)(1-u+u^2) = 1+u^3$$

shows why such cancellation would be invalid.

Instead, we extract the coefficient of a lattice monomial with multiplicity vector $\mathbf{m}$ from an independent-set product in a quantum torus. The common-history theorem applies to this coefficient before any blowup factorial is introduced. Its weights are positive Laurent series. A second extraction, this time of the squarefree *color* monomial $x_1 \cdots x_N$, forces their support to lie in the finite set $\{A, A-2, \ldots, -A\}$. Transposition of zero-one matrices then supplies the elementary expansion over $\mathbb{Z}_{\geq 0}[u]$.

Section **??** introduces the step-τ factorizations and their grading. Section **??** proves the static-statistic identity and the weighted coloring correspondence. Section **??** states the precise imported input, and Section **??** proves Theorem **??**. Section **??** completes the Dyck theorem and its classical specialization. Section **??** records direct consequences and examples. Appendix **??** gives the source correspondence at a fixed repository revision, and Appendix **??** describes independent exact computations. The connection to q,t-Catalan polynomials in [**?**] motivates the result; no additional Catalan identity is needed for the proofs below.

2. Step-τ factorizations and grading conventions

2.1. **Symmetric functions and coefficientwise positivity.** We work with the ring $\Lambda_{\mathbb{Z}}$ of symmetric functions in $\mathbf{x} = (x_1, x_2, \ldots)$. The elementary and complete homogeneous functions are

$$e_r(\mathbf{x}) = \sum_{a_1<\cdots<a_r} x_{a_1}\cdots x_{a_r}, \qquad h_r(\mathbf{x}) = \sum_{a_1\le\cdots\le a_r} x_{a_1}\cdots x_{a_r},$$

with $e_0 = h_0 = 1$. For a partition λ, write $e_\lambda = \prod_i e_{\lambda_i}$ and $h_\lambda = \prod_i h_{\lambda_i}$. The involution $\omega : \Lambda_{\mathbb{Z}} \to \Lambda_{\mathbb{Z}}$ is characterized by $\omega(e_r) = h_r$; equivalently, $\omega(s_\lambda) = s_{\lambda'}$. It acts coefficientwise on $\Lambda_{\mathbb{Z}}[u]$ and leaves u fixed.

An element $\sum_\lambda b_\lambda(u)e_\lambda$ is *coefficientwise e-positive* if every $b_\lambda(u) \in \mathbb{Z}_{\ge 0}[u]$. This is stronger than positivity after a numerical specialization of u. The analogous definition uses h_λ. Finite-variable restrictions will be denoted by a subscript r or by displaying $x_1, \ldots, x_r$. All our series have fixed homogeneous degree and are determined by these restrictions.

There are three distinct parameters in the proof. The integer N is the homogeneous degree in $\mathbf{x}$, the variable u records the inversion or Dyck statistic, and the Laurent variable v is used only in the quantum-torus argument. Its relation to u is $u = v^{-2}$, after a normalizing factor v^A. In particular, v is not a second Catalan parameter.

2.2. **Words and factorizations.** Let $\mathcal{W}(S)$ be the set of distinct rearrangements of the multiset S. Equal occurrences are not labeled. Fix $\tau \in \mathbb{Z}_{\ge 0}$.

Definition 2.1. A finite integer word $w = (w_1, \ldots, w_n)$ is an *affine step-τ word* if $w_{j+1} \le w_j + \tau$ for every $j < n$. It is a *dual step-τ word* if $w_{j+1} > w_j + \tau$ for every $j < n$. The empty word and each one-letter word satisfy both conditions.

No nonnegativity or initial-entry condition is imposed. These are conditions on individual factors, not on the multiset itself. A factorization of S is a sequence $\mathcal{F} = (F_1, F_2, \ldots)$ of finite words, all but finitely many empty, whose concatenation $w(\mathcal{F}) = F_1F_2\cdots$ lies in $\mathcal{W}(S)$. It is affine or dual when each factor has the corresponding property. Its monomial is

$$\mathbf{x}^{\mathcal{F}} = \prod_{a\ge 1} x_a^{|F_a|}. \tag{2.1}$$

Empty factors are important: factor indices are the color labels that will be used later. Our one-indexed convention differs from the zero-indexed convention in [**?**] only by renaming variables and factor indices.

2.3. **The statistic.** For integers a, b, put

$$d_\tau(a,b) = \begin{cases} \max(0, a+\tau-b), & a \le b, \\ \max(0, b+1+\tau-a), & a > b. \end{cases} \tag{2.2}$$

For a word w, define

$$\operatorname{dinv}_\tau(w) = \sum_{p<q} d_\tau(w_p, w_q). \tag{2.3}$$

The statistic of a factorization is that of its concatenation; there is no additional contribution attached to a factor boundary. Define

$$\mathcal{D}^*_{S,\tau}(\mathbf{x};u) = \sum_{\mathcal{F}\ \text{dual}} u^{\operatorname{dinv}_\tau(w(\mathcal{F}))}\mathbf{x}^{\mathcal{F}}, \tag{2.4}$$

$$\mathcal{D}_{S,\tau}(\mathbf{x};u) = \sum_{\mathcal{F}\ \text{affine}} u^{\operatorname{dinv}_\tau(w(\mathcal{F}))}\mathbf{x}^{\mathcal{F}}. \tag{2.5}$$

Their degree-d coefficients are

$$\mathrm{DS}^{*(\tau)}(S,d;\mathbf{x}) = [u^d]\mathcal{D}^*_{S,\tau}(\mathbf{x};u), \qquad \mathrm{DS}^{(\tau)}(S,d;\mathbf{x}) = [u^d]\mathcal{D}_{S,\tau}(\mathbf{x};u). \tag{2.6}$$

Both are understood to be zero for $d < 0$. Before symmetry has been established, we refer to (**??**)–(**??**) as factorization series.

Every coefficient of a fixed monomial is a finite sum: its exponents fix all factor lengths, and only finitely many rearrangements of S can be concatenations. An increasing relabeling of the used factor indices preserves concatenation and statistic, so the series are quasisymmetric. Their u-degrees are uniformly bounded for fixed S and τ, since there are finitely many words in $\mathcal{W}(S)$.

Remark 2.2. The integrality of the entries and of τ matters in the grading calculation below. We prove the full integer-step family; no assertion about arbitrary rational slopes is implicit in the terminology.

3. The weighted coloring model

3.1. **Separating the constant part of the statistic.** For $w \in \mathcal{W}(S)$, set

$$\mathrm{inv}_\tau(w) = \#\{(p,q) : p < q,\ 0 < w_p - w_q \le \tau\}. \tag{3.1}$$

Lemma 3.1 (Static decomposition)**.** *For every $w \in \mathcal{W}(S)$,*

$$\mathrm{dinv}_\tau(w) = \beta_\tau(S) + \mathrm{inv}_\tau(w). \tag{3.2}$$

Proof. Consider a pair of occurrences, independently of its order in the word. If both entries equal c_i, its contribution to (**??**) is τ. Summing over equal pairs gives $\tau \sum_i \binom{m_i}{2}$.

For two distinct values $c_i < c_j$, let $\delta = c_j - c_i$. If $\delta > \tau$, both orders contribute zero: for the decreasing order, $\tau + 1 - \delta \le 0$ because δ is an integer. If $1 \le \delta \le \tau$, then

$$d_\tau(c_i, c_j) = \tau - \delta, \qquad d_\tau(c_j, c_i) = \tau - \delta + 1.$$

There is an order-independent contribution $\tau - \delta$, together with an additional 1 exactly when the larger value precedes the smaller. There are $m_i m_j$ pairs with these two values. Summing over all value pairs proves (**??**). □

Corollary 3.2. *Every factorization of S has statistic in the interval*

$$\beta_\tau(S) \le \mathrm{dinv}_\tau \le \beta_\tau(S) + A_\tau(S).$$

Proof. Exactly $A_\tau(S)$ pairs of occurrences have distinct values within distance τ. Each contributes either 0 or 1 to inv_τ. □

The baseline is not just an equal-entry correction when $\tau > 1$: nearby distinct values also contribute to (**??**). This observation will account for the different grading shifts in the examples of Section **??**.

3.2. **The value graph and the coloring correspondence.** Define $G_{S,\tau}$ on the value levels $[\ell]$ by

$$ij \in E(G_{S,\tau}) \quad \Longleftrightarrow \quad i < j \text{ and } c_j - c_i \le \tau. \tag{3.3}$$

Vertex i has weight m_i.

Lemma 3.3. *The graph $G_{S,\tau}$ is a natural unit interval graph, with Hessenberg function*

$$h_{S,\tau}(i) = \max\{j \in [\ell] : c_j - c_i \le \tau\}. \tag{3.4}$$

Its weighted edge count is $A_\tau(S)$.

Proof. The maximum exists because $j = i$ is admissible. Increasing i increases the threshold $c_i + \tau$, so the maximum is weakly increasing. It is at least i. For $i < j$, condition (**??**) is exactly $j \le h_{S,\tau}(i)$. The assertion about the weighted edge count follows from (**??**). □

Proposition 3.4 (Dual factorizations as set colorings)**.** *There is a monomial-preserving bijection between dual step-τ factorizations of S and proper set colorings of $(G_{S,\tau}, \mathbf{m})$. Under this bijection,*

$$\mathrm{inv}_\tau(w(\mathcal{F})) = \mathrm{inv}_{G_{S,\tau}}(\gamma). \tag{3.5}$$

Consequently,

$$\mathcal{D}^*_{S,\tau}(\mathbf{x}; u) = u^{\beta_\tau(S)} C^-_{G_{S,\tau},\mathbf{m}}(\mathbf{x}; u). \tag{3.6}$$

Proof. A dual factor is strictly increasing, so it contains at most one copy of each value. All pairs of its values have distance greater than τ: this is immediate for consecutive values, and a nonconsecutive positive gap is even larger. Thus its value levels form an independent set of $G_{S,\tau}$.

Given $\mathcal{F} = (F_1, F_2, \ldots)$, set

$$\gamma_i = \{a \geq 1 : c_i \text{ occurs in } F_a\}.$$

There are exactly m_i such factors. Adjacent value levels cannot occur in the same factor, so γ is proper. Conversely, from a proper coloring, form F_a by listing, in increasing order, those c_i for which $a \in \gamma_i$. Its value levels are independent, and hence its successive gaps exceed τ. These constructions are inverse. They also give $|F_a| = \#\{i : a \in \gamma_i\}$, proving monomial preservation.

Consider $i < j$ with $c_j - c_i \leq \tau$, and choose colors $a \in \gamma_i$, $b \in \gamma_j$. Properness gives $a \neq b$. The occurrence of c_j precedes that of c_i in the concatenation precisely when $b < a$. Thus it contributes to (**??**) precisely when the pair (a, b) contributes to $\mathrm{inv}_{G_{S,\tau}}(\gamma)$. There is no contributing pair within a dual factor. This proves (**??**). Combining it with Lemma **??** proves (**??**). □

This identity uses inversions directly. Neither symmetry nor palindromicity has been used to reverse the grading.

3.3. **The clique blowup identity.** We record a proof of (**??**) to specify its exact normalization. Let $M_i = m_1 + \cdots + m_i$, with $M_0 = 0$. The blowup $\widetilde{G}$ has the consecutive blocks $\{M_{i-1} + 1, \ldots, M_i\}$. It is again a natural unit interval graph: its Hessenberg function is

$$\widetilde{h}(a) = M_{h(i)} \quad \text{for } M_{i-1} < a \leq M_i. \tag{3.7}$$

Proposition 3.5. *Let $X_{\widetilde{G}}$ use the usual ascent statistic on proper single-color assignments to $\widetilde{G}$. Then*

$$X_{\widetilde{G}}(\mathbf{x}; u) = \left(\prod_i [m_i]_u! \right) C^+_{G,\mathbf{m}}(\mathbf{x}; u). \tag{3.8}$$

The corresponding identity also holds with inversions on both sides.

Proof. A proper coloring of a clique uses distinct colors. Forgetting their order within each block produces a proper set coloring of G. For a fixed set coloring, the choices of order in different blocks are independent. All contributions from edges between blocks are fixed and equal the weighted ascent statistic. Within a block of size m_i, the generating polynomial of ascent pairs over all orders of its colors is $[m_i]_u!$. Indeed, the inversion generating polynomial of permutations has the recurrence $P_m = [m]_u P_{m-1}$, obtained by inserting the largest entry, and reversing the order exchanges ascent and inversion pairs. Multiplying the block contributions gives (**??**). The inversion version has the same proof. □

The proposition explains the relation to ordinary chromatic functions, but it is not a positivity argument for the quotient. We now work with the weighted target itself.

4. The positive common-history input

This section specifies the external input from [**?**, **?**]. We do not reproduce the wall-crossing and common-history constructions. Instead, we state exactly the properties used below and identify the source declarations in Appendix **??**. In particular, the theorem we import is not merely the final elementary-positivity theorem for unweighted graphs.

4.1. **Quantum-torus notation.** Let $K = \mathbb{Q}((v))$, and let L be a finite-rank free abelian group with an integral alternating bilinear form Ω. Its quantum torus over K is the vector space with basis $\{T^a : a \in L\}$ and multiplication

$$T^a T^b = v^{-\Omega(a,b)} T^{a+b}. \tag{4.1}$$

Only finite linear combinations of basis monomials are needed here; their coefficients may be Laurent series. The coefficient of T^R in F is denoted by $[T^R]F$. Bilinearity gives

$$T^{a_1} \cdots T^{a_r} = v^{-\sum_{p<q} \Omega(a_p,a_q)} T^{a_1+\cdots+a_r}. \tag{4.2}$$

The source uses a unit called `LaurentRay.vUnit`. Its kth power has Laurent exponent $-k$, so that unit is v^{-1} in our notation. This explains the sign in (**??**); see Appendix **??**.

Let V be a finite alphabet and let $\iota : \mathbb{Z}^V \hookrightarrow L$ be an injective homomorphism. If ϵ_a is its standard coordinate vector, set $u_a = \iota(\epsilon_a)$. A subset $I \subseteq V$ is *independent* if $\Omega(u_a, u_b) = 0$ for all $a, b \in I$. Define

$$\mathscr{E}_k = \sum_{\substack{I \subseteq V \text{ independent} \\ |I|=k}} T^{\sum_{a\in I} u_a}, \qquad \mathscr{E}_0 = 1. \tag{4.3}$$

In particular, $\mathscr{E}_k = 0$ for $k > |V|$. These are the independent-set elements of the source, with the displayed normalization. A contributing set occurs with coefficient 1; there is no normalization by $[k]_u!$ or by $k!$.

4.2. **Triangular realizations.** The triangular construction in [**?**] supplies a finite alphabet $V = V_D$, a lattice $L = L_D$, an alternating form Ω, an injective coordinate map ι, an anchor $a_0 \in L$, and a finite list of root vectors $\rho_j \in L$, $j \in J$. Write

$$\mathscr{C} = \left\{ \sum_{j\in J} n_j \rho_j : n_j \in \mathbb{Z}_{\geq 0} \right\}. \tag{4.4}$$

Thus membership in $\mathscr{C}$ means a nonnegative *integral* combination, not just membership in the real cone spanned by the roots.

Proposition 4.1 (Triangular graph realization, OpenAI)**.** *For every natural unit interval graph G on $[\ell]$, one can choose the triangular data above and an injection $b : [\ell] \hookrightarrow V$ such that*

$$\Omega(u_{b(i)}, u_{b(j)}) = -\mathbf{1}_{\{ij \in E(G)\}} \qquad (i < j). \tag{4.5}$$

The coordinate map ι is injective on all of $\mathbb{Z}^V$, including coordinates outside the embedded graph.

This is the combination of the source's `graph_triangular_embedding`, `graphVertex_pair`, and `includeVertices_injective`. We always choose the actual triangular realization supplied by that construction. The next theorem is asserted for those data, not for arbitrary lattice data satisfying only (**??**).

Theorem 4.2 (Positive common histories, OpenAI)**.** *Fix such a triangular realization. Let $N \in \mathbb{Z}_{\geq 0}$ and $R \in L$ satisfy*

$$R - N a_0 \in \mathscr{C}. \tag{4.6}$$

There are a finite set $\mathcal{H}$, profiles $\mu(\eta) = (\mu_0(\eta), \ldots, \mu_D(\eta)) \in \mathbb{Z}_{\geq 0}^{D+1}$, and weights $W_\eta(v) \in \mathbb{Q}((v))$ such that

$$\sum_{j=0}^{D} \mu_j(\eta) = N, \qquad [v^s] W_\eta(v) \in \mathbb{Z}_{\geq 0} \quad (s \in \mathbb{Z}). \tag{4.7}$$

For every $r \geq 1$ and every $\alpha = (\alpha_1, \ldots, \alpha_r) \in \mathbb{Z}^r_{\geq 0}$ with $\sum_a \alpha_a = N$,

$$[T^R]\mathscr{E}_{\alpha_1} \cdots \mathscr{E}_{\alpha_r} = \sum_{\eta \in \mathcal{H}} W_\eta(v)[\mathbf{z}^{\mu(\eta)}] \prod_{a=1}^{r} e_{\alpha_a}(z_0, \ldots, z_D). \tag{4.8}$$

The same set, profiles, and weights work simultaneously for all r and α.

Here $[v^s]$ is the ordinary Laurent coefficient, allowed at any integer exponent. Theorem **??** initially does *not* assert that the weights are Laurent polynomials, nor that their exponents have a prescribed parity. Both facts will be proved for our targets in Section **??**.

Identification with the cited theorem. The source's `triangular_common_history` is quantified over every homogeneous elementary expression of degree N. Take the expression whose ordered sequence of elementary indices is $(\alpha_1, \ldots, \alpha_r)$. Its torus evaluation is the product on the left of (**??**), and its ordinary evaluation is the product on the right. This is the same specialization used in the source's unweighted `graph_common_history` proof.

For a representation $R - Na_0 = \sum_j n_j \rho_j$, the source's root degree is $d = \sum_j n_j$. Thus (**??**) provides exactly its hypothesis `HasRootDegree`, with no squarefreeness condition on R. The finite index set in (**??**) consists of the anchored leaves of the source's common history. Their profiles have total degree N; in fact they are weakly decreasing, a property we will not need. Finiteness follows from `leafFintype`, and `leafWeight_positive` gives positivity of every weight. Finally, the source defines `LaurentPositive.Positive` by requiring each coefficient to equal a natural number. It therefore gives exactly the integer positivity in (**??**). The common-history theorem chooses its history before quantifying over elementary expressions, which gives the required uniformity in r and α. □

The product form (**??**) is enough for the rest of the paper. No assertion about commutativity of arbitrary torus elements is needed: the products on the left retain the displayed order.

5. Multiplicity targets and weighted elementary positivity

For $N = 0$, both weighted series are 1, so Theorem **??** is immediate. Fix G and $\mathbf{m}$ with $N > 0$, and choose the realization from Proposition **??**. All history data in this section are attached to this one realization and one target.

5.1. **The root-cone hypothesis for multiplicities.** Set

$$R_{\mathbf{m}} = \sum_{i=1}^{\ell} m_i u_{b(i)}. \tag{5.1}$$

Lemma 5.1 (Admissibility of the multiplicity target)**.** *The target $R_{\mathbf{m}}$ satisfies*

$$R_{\mathbf{m}} - Na_0 \in \mathscr{C}. \tag{5.2}$$

Proof. Apply the source's `squarefree_rootDegree` to the injection from a one-element set to the triangular vertex $b(i)$. Its conclusion is

$$u_{b(i)} - a_0 = \sum_{j \in J} n_{ij} \rho_j \quad \text{with } n_{ij} \in \mathbb{Z}_{\geq 0}.$$

This is the singleton case of the cited result; it imposes no restriction on how often that singleton may be used in a subsequent sum. Therefore

$$R_{\mathbf{m}} - Na_0 = \sum_i m_i (u_{b(i)} - a_0) = \sum_{j \in J} \left(\sum_i m_i n_{ij} \right) \rho_j \in \mathscr{C}.$$

More explicitly, if the singleton root degree is $d_i = \sum_j n_{ij}$, then the displayed expression has root degree $\sum_i m_i d_i$. This verifies the precise hypothesis of Theorem **??**. □

The use of a squarefree *singleton* result in this lemma should not be confused with a squarefreeness restriction on the final target. The coefficients m_i in (**??**) are arbitrary positive integers.

5.2. **The weighted coefficient identity.** For $r \geq 1$, let C_r^- be the restriction of $C_{G,\mathbf{m}}^-$ to colors in $[r]$. Recall the weighted edge count A from (**??**).

Lemma 5.2 (Weighted coefficient extraction)**.** *For $\alpha \in \mathbb{Z}_{\geq 0}^r$ with $\sum_a \alpha_a = N$,*

$$[T^{R_{\mathbf{m}}}]\mathscr{E}_{\alpha_1} \cdots \mathscr{E}_{\alpha_r} = v^A[\mathbf{x}^\alpha]C_r^-(\mathbf{x}; v^{-2}). \tag{5.3}$$

Proof. Expanding (**??**) chooses an independent subset $I_a \subseteq V$ of cardinality α_a for each color $a \in [r]$. Its total lattice exponent is

$$\sum_{a=1}^{r} \sum_{z \in I_a} u_z.$$

Let $n_z = \#\{a : z \in I_a\}$. Since $u_z = \iota(\epsilon_z)$ and ι is injective, this exponent equals $R_{\mathbf{m}}$ if and only if

$$n_z = \begin{cases} m_i, & z = b(i), \\ 0, & z \notin b([\ell]). \end{cases} \tag{5.4}$$

All n_z are nonnegative, so no occurrence of an outside vertex can cancel another coordinate.

For a contributing tuple, define

$$\gamma_i = \{a \in [r] : b(i) \in I_a\}. \tag{5.5}$$

Equation (**??**) gives $|\gamma_i| = m_i$. By (**??**), independence of I_a is equivalent to properness for vertices receiving color a. Thus γ is a proper set coloring, and $|I_a| = \alpha_a$ says that its monomial is $\mathbf{x}^\alpha$. Conversely, such a coloring gives $I_a = \{b(i) : a \in \gamma_i\}$. This is a bijection, including when some $\alpha_a = 0$. In particular, it does not order or label the m_i occurrences at vertex i; no factorial factor appears.

It remains to compute the torus weight. Write $v_a^{\mathrm{lat}} = \sum_{z \in I_a} u_z$ for the lattice exponent of the ath factor, to distinguish it from the Laurent variable v. By (**??**), the coefficient contributed by the tuple is

$$v^{-\sum_{a<b} \Omega(v_a^{\mathrm{lat}}, v_b^{\mathrm{lat}})}. \tag{5.6}$$

Consider an edge $i < j$ and a pair of colors $a \in \gamma_i$, $b \in \gamma_j$. Properness excludes $a = b$. If $a < b$, the ordered pairing is $\Omega(u_{b(i)}, u_{b(j)}) = -1$, contributing $+1$ to the exponent in (**??**). If $a > b$, the earlier factor contributes $u_{b(j)}$ and the later one contributes $u_{b(i)}$; the pairing is $+1$, contributing -1 to that exponent. Hence the edge contributes

$$m_i m_j - 2\#\{(a, b) \in \gamma_i \times \gamma_j : a > b\}.$$

A nonedge contributes zero by (**??**). Two occurrences of the same vertex also contribute zero because Ω is alternating. Summing over all edges gives $A - 2\,\mathrm{inv}_G(\gamma)$. Thus the term associated to γ has weight $v^A(v^{-2})^{\mathrm{inv}_G(\gamma)}$, proving (**??**). □

The lemma is the new interpretation of coefficient extraction required for multiplicities. The common-history theorem itself is unchanged.

5.3. **From common histories to elementary products.** Apply Theorem **??** at $R_{\mathbf{m}}$, using Lemma **??**. Fix the resulting $\mathcal{H}$, $\mu(\eta)$, and $W_\eta(v)$. These do not depend on the number of colors.

Lemma 5.3 (Zero-one matrix transposition)**.** *If $\alpha \in \mathbb{Z}_{\geq 0}^r$ and $\mu \in \mathbb{Z}_{\geq 0}^{D+1}$ have the same total size, then*

$$[\mathbf{z}^\mu] \prod_{a=1}^{r} e_{\alpha_a}(z_0, \ldots, z_D) = [\mathbf{x}^\alpha] \prod_{j=0}^{D} e_{\mu_j}(x_1, \ldots, x_r). \tag{5.7}$$

Proof. On the left, choose a subset of $\{0,\dots,D\}$ of size α_a for each factor indexed by a. Its indicator vectors form an $r\times(D+1)$ zero-one matrix with row sums α and column sums μ. On the right, the choices give the transpose matrices, with column sums α and row sums μ. Transposition is a bijection. Zero entries in either profile cause no change to the argument. □

For each history η, let $\lambda(\eta)\vdash N$ be the partition obtained by sorting the nonzero entries of $\mu(\eta)$. Combining (**??**), (**??**), and (**??**), for every α of total degree N, gives

$$\begin{aligned} v^A[\mathbf{x}^\alpha]C_r^-(\mathbf{x};v^{-2}) &= \sum_{\eta\in\mathcal{H}} W_\eta(v)[\mathbf{z}^{\mu(\eta)}]\prod_{a=1}^{r} e_{\alpha_a}(\mathbf{z}) \\ &= \sum_{\eta\in\mathcal{H}} W_\eta(v)[\mathbf{x}^\alpha]e_{\lambda(\eta)}(x_1,\dots,x_r). \end{aligned}$$

Both sides concern homogeneous polynomials of degree N in the color variables. Equality of all coefficients therefore yields

$$v^A C_r^-(\mathbf{x};v^{-2}) = \sum_{\eta\in\mathcal{H}} W_\eta(v)e_{\lambda(\eta)}(x_1,\dots,x_r). \tag{5.8}$$

This already gives symmetry over K, but we still need an expansion over $\mathbb{Z}_{\ge0}[u]$, not merely over positive Laurent series.

5.4. **A positive test coefficient forces finite support.** For a Laurent series $F(v)$, let $\operatorname{supp}_v(F)=\{s\in\mathbb{Z}:[v^s]F\neq0\}$.

Lemma 5.4 (Support of a positive sum)**.** *Let $F_1,\dots,F_t\in\mathbb{Q}((v))$ have nonnegative coefficients, and let $c_1,\dots,c_t$ be strictly positive rational numbers. If $\sum_i c_iF_i$ is supported on a finite set $E\subseteq\mathbb{Z}$, then $\operatorname{supp}_v(F_i)\subseteq E$ for every i.*

Proof. For $s\notin E$, coefficient extraction gives $0=\sum_i c_i[v^s]F_i$. Each summand is nonnegative. Since each $c_i>0$, all the coefficients $[v^s]F_i$ are zero. □

Lemma 5.5 (Support and parity of history weights)**.** *For every $\eta\in\mathcal{H}$,*

$$\operatorname{supp}_v(W_\eta)\subseteq E_A := \{A-2k : 0\le k\le A\}. \tag{5.9}$$

Consequently there is a polynomial $P_\eta(u)\in\mathbb{Z}_{\ge0}[u]$ of degree at most A such that

$$W_\eta(v) = v^A P_\eta(v^{-2}). \tag{5.10}$$

Proof. Take $r=N$ in (**??**) and extract the coefficient of $x_1\cdots x_N$. Because $\sum_j\mu_j(\eta)=N$, the coefficient in the elementary product is

$$M_\eta = [x_1\cdots x_N]\prod_{j=0}^{D} e_{\mu_j(\eta)}(x_1,\dots,x_N) = \frac{N!}{\prod_{j=0}^{D}\mu_j(\eta)!} > 0. \tag{5.11}$$

Indeed, it counts ordered disjoint subsets of the N colors with the specified cardinalities. Thus

$$\sum_{\substack{\gamma \text{ proper in } [N]\\ \mathbf{x}^\gamma = x_1\cdots x_N}} v^{A-2\operatorname{inv}_G(\gamma)} = \sum_{\eta\in\mathcal{H}} M_\eta W_\eta(v). \tag{5.12}$$

The left side is a finite Laurent polynomial. Since an edge contributes between 0 and m_im_j inversions, $0\le\operatorname{inv}_G(\gamma)\le A$, and its support lies in E_A. All Laurent coefficients of each W_η are nonnegative integers by Theorem **??**. Apply Lemma **??**, using the strict positivity of (**??**), to obtain (**??**).

Set

$$P_\eta(u) = \sum_{k=0}^{A}[v^{A-2k}]W_\eta(v)\,u^k. \tag{5.13}$$

Its coefficients are nonnegative integers. The support statement says that these are all the coefficients of W_η, which proves (**??**). □

The test coefficient in this argument is squarefree in the *color* variables. It does not make the lattice target $R_{\mathbf{m}}$ squarefree: a vertex of weight m_i still receives m_i distinct colors. The two coefficient extractions serve different purposes and are compatible. The same positive-support principle occurs in the unweighted source argument; the proof above gives its complete weighted form.

5.5. **Completion of the weighted theorem.** Substitute (**??**) into (**??**) and cancel the invertible monomial v^A:

$$C_r^-(\mathbf{x}; v^{-2}) = \sum_{\eta \in \mathcal{H}} P_\eta(v^{-2}) e_{\lambda(\eta)}(x_1, \dots, x_r).$$

The substitution homomorphism $\mathbb{Z}[u] \to \mathbb{Q}((v))$, $u \mapsto v^{-2}$, is injective: distinct powers of u map to distinct Laurent powers. Coefficientwise in the color variables, the preceding identity gives

$$C_r^-(\mathbf{x}; u) = \sum_{\eta \in \mathcal{H}} P_\eta(u) e_{\lambda(\eta)}(x_1, \dots, x_r). \tag{5.14}$$

The right side uses the same history polynomials for every r. The restrictions are compatible under $x_r = 0$, so passage to symmetric functions gives

$$C_{G,\mathbf{m}}^-(\mathbf{x}; u) = \sum_{\lambda \vdash N} B_{G,\mathbf{m},\lambda}(u) e_\lambda(\mathbf{x}), \tag{5.15}$$

where

$$B_{G,\mathbf{m},\lambda}(u) = \sum_{\substack{\eta \in \mathcal{H} \\ \lambda(\eta) = \lambda}} \sum_{k=0}^{A} [v^{A-2k}] W_\eta(v)\, u^k \in \mathbb{Z}_{\geq 0}[u]. \tag{5.16}$$

This proves elementary positivity, integrality, symmetry, and the degree bound. The history family is not asserted to be canonical, but the polynomials $B_{G,\mathbf{m},\lambda}$ are canonical by uniqueness of the elementary expansion. Formula (**??**) is a positive history formula, not a claim of a new tableau or permutation formula for these coefficients.

It remains to relate the orientations and prove palindromicity. For a proper set coloring in $[r]$, replace every color a by $r + 1 - a$. This is an involution that interchanges ascents and inversions and reverses the color variables. Hence

$$C_r^+(x_1, \dots, x_r; u) = C_r^-(x_r, \dots, x_1; u).$$

The symmetry just proved makes the right side equal to $C_r^-(x_1, \dots, x_r; u)$. Thus $C^+ = C^-$ in the stable limit. On each edge, equality of colors is prohibited, so

$$\mathrm{asc}_G(\gamma) + \mathrm{inv}_G(\gamma) = A. \tag{5.17}$$

Termwise, it follows that

$$C_{G,\mathbf{m}}^+(\mathbf{x}; u) = u^A C_{G,\mathbf{m}}^-(\mathbf{x}; u^{-1}).$$

Comparing elementary coefficients in (**??**) now proves (**??**). This completes the proof of Theorem **??**.

Remark 5.6. The only factor cancelled in the positivity argument is the monomial v^A. No quantum factorial, ordinary factorial, or polynomial with more than one nonzero coefficient is cancelled from a positive expansion. The factorials in (**??**) are merely the strictly positive scalar test coefficients; they are not used to divide history weights.

6. Dyck positivity and the classical specialization

6.1. **The dual series.** Apply Theorem **??** to the natural unit interval graph $G_{S,\tau}$ of Lemma **??**, with weights $\mathbf{m} = (m_1, \dots, m_\ell)$. Its weighted edge count is $A_\tau(S)$. By Proposition **??**,

$$\mathcal{D}^*_{S,\tau}(\mathbf{x}; u) = u^{\beta_\tau(S)} \sum_{\lambda \vdash N} B_{G_{S,\tau},\mathbf{m},\lambda}(u) e_\lambda(\mathbf{x}). \tag{6.1}$$

This proves symmetry and the dual expansion in Theorem **??**. Explicitly, we may take

$$a_{\lambda,k}(S,\tau) = [u^k] B_{G_{S,\tau},\mathbf{m},\lambda}(u) = \sum_{\substack{\eta \in \mathcal{H} \\ \lambda(\eta) = \lambda}} [v^{A_\tau(S)-2k}] W_\eta(v). \tag{6.2}$$

The histories here are chosen for $G_{S,\tau}$ and $R_{\mathbf{m}}$ as in Section **??**. Formula (**??**) gives nonnegative integers and proves the asserted support and palindromicity of these coefficients.

6.2. **Complementary factor boundaries.** To pass to the affine series, we prove the involution relation for all integer steps. This also avoids assuming affine symmetry in advance. The argument is the fixed-word complement argument appearing in the step-one setting in [**?**, Corollary 3.20].

For $D \subseteq [N-1]$, define the fundamental quasisymmetric function

$$F_{N,D}(\mathbf{x}) = \sum_{\substack{1 \le a_1 \le \cdots \le a_N \\ j \in D \Rightarrow a_j < a_{j+1}}} x_{a_1} \cdots x_{a_N}. \tag{6.3}$$

These functions form a basis of the degree-N quasisymmetric functions. Let ψ be the degreewise linear involution determined by

$$\psi(F_{N,D}) = F_{N,[N-1]\setminus D}. \tag{6.4}$$

On degree zero put $\psi(1) = 1$.

Lemma 6.1. *The map ψ preserves symmetric functions and restricts there to ω.*

Proof. For a standard Young tableau T in English convention, let $\mathrm{Des}(T)$ consist of the integers j for which $j+1$ lies in a lower row than j. Standardization of semistandard tableaux gives

$$s_\lambda = \sum_{T \in \mathrm{SYT}(\lambda)} F_{N,\mathrm{Des}(T)}. \tag{6.5}$$

In this standardization, entries of the same value are numbered from left to right, so equality is allowed exactly away from tableau descents; this gives (**??**) directly from the semistandard generating function. For consecutive entries in a standard tableau, transposition complements the descent set:

$$\mathrm{Des}(T^{\mathsf{t}}) = [N-1] \setminus \mathrm{Des}(T).$$

Indeed, if $j+1$ lies in a lower row, it cannot lie strictly to the right of j: a cell on a row–column path between the two would have to carry an integer strictly between j and $j+1$. If it does not lie in a lower row, it must lie to the right. Applying (**??**) to (**??**) therefore gives $\psi(s_\lambda) = s_{\lambda'} = \omega(s_\lambda)$. Since Schur functions form a basis of $\Lambda_{\mathbb{Z}}$, the assertion follows. □

Proposition 6.2 (The affine–dual involution)**.** *For every S and integer $\tau \ge 0$,*

$$\mathcal{D}_{S,\tau}(\mathbf{x}; u) = \omega \mathcal{D}^*_{S,\tau}(\mathbf{x}; u). \tag{6.6}$$

The identity holds separately in each degree of u.

Proof. Fix $w \in \mathcal{W}(S)$ and write

$$D^*_\tau(w) = \{j \in [N-1] : w_{j+1} \le w_j + \tau\},$$
$$D_\tau(w) = \{j \in [N-1] : w_{j+1} > w_j + \tau\}.$$

A factorization with concatenation w is equivalently a weakly increasing sequence $(a_1, \dots, a_N)$ of factor labels. Whenever $a_j = a_{j+1}$, the pair (w_j, w_{j+1}) lies within one factor and must satisfy its local inequality. Thus dual factorizations force a strict increase exactly at the positions in $D^*_\tau(w)$, while affine factorizations force one exactly at the positions in $D_\tau(w)$. Additional strict increases are allowed in either case. Empty factors correspond to skipped labels, so no further choices or multiplicities are missing. The fixed-word generating functions are consequently $F_{N,D^*_\tau(w)}$ and $F_{N,D_\tau(w)}$.

The two boundary sets are complements. Since the exponent $\mathrm{dinv}_\tau(w)$ depends on the concatenated word alone, we obtain

$$\mathcal{D}^*_{S,\tau}(\mathbf{x}; u) = \sum_{w \in \mathcal{W}(S)} u^{\mathrm{dinv}_\tau(w)} F_{N,D^*_\tau(w)}(\mathbf{x}),$$
$$\mathcal{D}_{S,\tau}(\mathbf{x}; u) = \sum_{w \in \mathcal{W}(S)} u^{\mathrm{dinv}_\tau(w)} F_{N,D_\tau(w)}(\mathbf{x}) = \psi \mathcal{D}^*_{S,\tau}(\mathbf{x}; u).$$

The dual series is symmetric by (**??**), so Lemma **??** replaces ψ by ω. This proves (**??**), including its coefficientwise version. For $S = \varnothing$, both sides are 1. □

Applying $\omega(e_\lambda) = h_\lambda$ to (**??**) gives the affine expansion with precisely the same integers (**??**). This completes the proof of Theorem **??**.

6.3. **Hawkes's original functions.** We now identify the step-one specialization with the exact functions in [**?**]. In Definition 2.1 of that paper, the statistic is

$$\mathrm{di}(w) = \#\{p < q : w_p = w_q + 1\}. \tag{6.7}$$

Definitions 2.2–2.5 use the affine and dual factor inequalities of Definition **??** at $\tau = 1$. In particular, equal pairs are not counted by (**??**). Put

$$\kappa(S) = \sum_i \binom{m_i}{2}. \tag{6.8}$$

Since all entries are integers, $0 < w_p - w_q \le 1$ is equivalent to $w_p = w_q + 1$. Also, the distinct-value terms in $\beta_1(S)$ all vanish. Lemma **??** therefore gives

$$\beta_1(S) = \kappa(S), \qquad \mathrm{dinv}_1(w) = \kappa(S) + \mathrm{di}(w). \tag{6.9}$$

After renaming Hawkes's variable x_{a-1} as our x_a, the fixed-degree identifications are

$$\mathrm{DS}^*_{\mathrm{H}}(S, d; \mathbf{x}) = \mathrm{DS}^{*(1)}(S, d + \kappa(S); \mathbf{x}), \tag{6.10}$$
$$\mathrm{DS}_{\mathrm{H}}(S, d; \mathbf{x}) = \mathrm{DS}^{(1)}(S, d + \kappa(S); \mathbf{x}), \tag{6.11}$$

where the subscript H denotes the original notation in [**?**, Definition 2.5], not a different family.

Corollary 6.3 (Positivity of the original Dyck functions)**.** *For every finite multiset S and every $d \ge 0$,*

$$\mathrm{DS}^*_{\mathrm{H}}(S, d; \mathbf{x}) = \sum_{\lambda \vdash N} a_{\lambda,d}(S, 1) e_\lambda(\mathbf{x}), \qquad \mathrm{DS}_{\mathrm{H}}(S, d; \mathbf{x}) = \sum_{\lambda \vdash N} a_{\lambda,d}(S, 1) h_\lambda(\mathbf{x}), \tag{6.12}$$

where coefficients outside $0 \le d \le A_1(S)$ are zero. All these coefficients are nonnegative integers, and $a_{\lambda,d}(S, 1) = a_{\lambda,A_1(S)-d}(S, 1)$.

Proof. Insert (**??**)–(**??**) into Theorem **??** and use $\beta_1(S) = \kappa(S)$. □

Thus the original generating series equal $u^{-\kappa(S)}\mathcal{D}^*_{S,1}$ and $u^{-\kappa(S)}\mathcal{D}_{S,1}$. These expressions are polynomials in u by the established lower degree bound. The removal of this monomial shift is unrelated to, and does not entail, division by a quantum factorial.

7. Consequences and examples

7.1. **Generalized Hessenberg chromatic functions.** Theorem **??** also has a direct formulation in the language of [**?**]. Let

$$P = \{M_1 < \cdots < M_\ell = N\}, \qquad M_0 = 0,$$

and let $H : P \to P$ be weakly increasing and extensive. This is a generalized Hessenberg function. Define

$$m_i = M_i - M_{i-1}, \qquad H(M_i) = M_{h(i)}. \tag{7.1}$$

Then $h : [\ell] \to [\ell]$ is weakly increasing and extensive. Conversely, any h and positive weights determine P and H in this way. The weighted graph in [**?**, Definitions 4.1–4.3] joins M_i to M_j for $M_i < M_j \leq H(M_i)$ and gives M_i the weight m_i. Its coloring sets, monomial, and ascent count are exactly those defining $C^+_{G,\mathbf{m}}$. Consequently,

$$\mathsf{csf}_u(H) = C^+_{G,\mathbf{m}}(\mathbf{x}; u). \tag{7.2}$$

Corollary 7.1. *The generalized Hessenberg chromatic function $\mathsf{csf}_u(H)$ is coefficientwise e-positive for every generalized Hessenberg function H. The equivariant Poincaré polynomial $\mathcal{F}(H)$ in the normalization of* [**?**] *is coefficientwise h-positive.*

Proof. The first assertion follows from (**??**) and Theorem **??**. The identity $\mathsf{csf}_u(H) = \omega\mathcal{F}(H)$ of [**?**, Theorem 4.6] gives the second. □

This answers the positivity question posed in [**?**, Question 4.9], with the common-history input specified in Section **??**. No geometric result beyond the cited character identity is needed for this consequence.

7.2. **Boundary cases.** For the empty multiset, both series are 1. For $\tau = 0$, the value graph has no edges and both $A_0(S)$ and $\beta_0(S)$ vanish. Each vertex chooses its color set independently, giving

$$\mathcal{D}^*_{S,0} = \prod_i e_{m_i}, \qquad \mathcal{D}_{S,0} = \prod_i h_{m_i}. \tag{7.3}$$

For a single value repeated m times, the graph also has no edges, but the equal-pair shift remains:

$$\mathcal{D}^*_{c^m,\tau} = u^{\tau\binom{m}{2}} e_m, \qquad \mathcal{D}_{c^m,\tau} = u^{\tau\binom{m}{2}} h_m. \tag{7.4}$$

These formulas in particular distinguish the shift from the weighted-edge statistic.

At the opposite extreme, if $c_\ell - c_1 \leq \tau$, the value graph is complete. Every color used in a proper set coloring is then used once. Order these N colors increasingly, and record the vertex receiving each one. The resulting word has content $1^{m_1} \cdots \ell^{m_\ell}$, and its ordinary inversion number equals $\mathrm{inv}_G(\gamma)$. Hence

$$C^-_{G,\mathbf{m}}(\mathbf{x}; u) = \begin{bmatrix} N \\ m_1, \ldots, m_\ell \end{bmatrix}_u e_N(\mathbf{x}), \tag{7.5}$$

where the Gaussian multinomial is defined here by the inversion generating polynomial of these multiset words. This directly proves its nonnegative coefficients. Its familiar factorial quotient is an identity for that polynomial, not a positivity inference used in our proof.

7.3. **A repeated-value example at two steps.** Let

$$S = \{0, 0, 1, 2\}, \qquad \mathbf{m} = (2, 1, 1), \qquad N = 4.$$

At $\tau = 1$, the value graph is the path 0–1–2, with $h = (2, 3, 3)$, $A_1(S) = 3$, and $\beta_1(S) = 1$. Its weighted function is

$$C^-_{G,\mathbf{m}} = (1 + u + u^2 + u^3)e_4 + (u + u^2)e_{3,1}. \tag{7.6}$$

For a direct check, the only monomial types are $(2, 1, 1)$ and $(1, 1, 1, 1)$. Enumerating proper color sets gives

$$[x_1^2 x_2 x_3]C^- = u + u^2, \qquad [x_1 x_2 x_3 x_4]C^- = 1 + 5u + 5u^2 + u^3.$$

Now $e_{3,1} = m_{2,1,1} + 4m_{1,1,1,1}$ and $e_4 = m_{1,1,1,1}$, which gives (**??**). Symmetry is known from Theorem **??**; alternatively, one may check each ordered placement of the squared variable directly. The step-one dual series is

$$\mathcal{D}^*_{S,1} = ue_4 + u^2(e_4 + e_{3,1}) + u^3(e_4 + e_{3,1}) + u^4 e_4. \tag{7.7}$$

The affine series replaces every e by h. Hawkes's original series in this example are obtained by removing the leading factor u, since $\kappa(S) = 1$.

At $\tau = 2$, the additional edge between the levels 0 and 2 makes the graph complete. We have

$$A_2(S) = 5, \qquad \beta_2(S) = 2 + 2 + 1 = 5.$$

In the latter expression the three terms come from the equal pair, the pairs of values $0, 1$, and the pair of values $1, 2$. Equation (**??**) gives

$$C^-_{G,\mathbf{m}} = [3]_u[4]_u e_4 = (1 + 2u + 3u^2 + 3u^3 + 2u^4 + u^5)e_4, \tag{7.8}$$

$$\mathcal{D}^*_{S,2} = u^5(1 + 2u + 3u^2 + 3u^3 + 2u^4 + u^5)e_4. \tag{7.9}$$

Again the affine series has the same coefficient polynomial with h_4 in place of e_4. This example shows why the step-τ value graph must include every pair of levels at distance at most τ, not just consecutive occupied levels.

7.4. **Scope of the conclusions.** The history formula proves nonnegative integrality and the argument with color reversal proves palindromicity. Neither assertion alone implies unimodality or log-concavity of the elementary coefficient polynomials. In particular, no answer to the stronger log-concavity question in [**?**, Question 4.10] is claimed here. Likewise, the positivity theorem applies to Hawkes's functions used in q, t-Catalan theory, but it does not by itself assert a new coefficient formula for a separate Catalan specialization. The main new mechanism is the multiplicity target in the common-history identity, together with the positive finite-support extraction.

Appendix A. Correspondence with the cited formalization

The external input is cited at the fixed revision `fd4aeeb` of the OpenAI mathematics repository [**?**]. The manuscript [**?**] supplies the broader argument; the exact intermediate results used here are identified below by module and declaration. All paths are relative to `lean/OAI/Combinatorics/Chromatic/`. The displayed module names are hyperlinks to that revision, rather than to a mutable branch.

A.1. **Graph realization and the multiplicity target.** The module `Walls/GraphTriangularEmbedding.lean` contains `graph_triangular_embedding`. It embeds the natural unit interval graph in a triangular alphabet with the prescribed pairings. The passage to the extended lattice and the negative-edge sign are expressed by `graphVertex_pair` in `Histories/GraphCommonHistory.lean`. The vertex-coordinate inclusion is injective; the source uses `includeVertices_injective` when extracting graph coefficients. This is the inclusion of the vertex summand into the extended lattice, so no linear relation among the vertex-coordinate vectors is introduced by the anchor coordinates.

The same graph-history module contains `squarefree_rootDegree`. Its domain is an arbitrary injection from a finite set to the triangular vertices. Taking the domain to have one element gives $u_{b(i)} - a_0 \in \mathscr{C}$. Lemma **??** then takes an integral linear combination with coefficients m_i. The source's theorem need not be modified or applied to a noninjective map. This singleton use is the exact justification for the multiplicity target.

A.2. **Independent elements and the Laurent parameter.** The definitions `IsIndependent`, `setMonomial`, and `independentElement` are in `QuantumTorus/IndependentSetElements.lean`. They give precisely the unnormalized independent-set sum (**??**). In `Walls/HNTorusFactor.lean`, `LaurentRay.vUnit_zpow` identifies the kth power of the parameter unit with the Hahn monomial at exponent $-k$. Thus the source's parameter unit is our v^{-1}, and a torus pairing $\Omega(a, b)$ contributes $v^{-\Omega(a,b)}$. This sign convention is responsible for $v^{A-2\,\mathrm{inv}}$ rather than $v^{2\,\mathrm{inv}-A}$ in Lemma **??**.

A.3. **The common history and its positivity.** The central declaration is `triangular_common_history` in `Histories/TriangularCommonHistory.lean`. Its parameters include a homogeneous degree, a root degree, and an arbitrary lattice target satisfying `HasRootDegree` after subtracting the anchor base. Its conclusion chooses one history and then quantifies over homogeneous elementary expressions. The ordered-product specialization in Theorem **??** uses `ElementaryExpr.sequence`; the same specialization is visible in the proof of `graph_common_history` in the graph-history module. Nothing in this step imposes a squarefree target.

The histories end at anchored leaves. Their profile degree is recorded by `leafProfile_degree`; the dominant-profile property is stronger than the nonnegativity and total-degree conditions we need. In `Histories/CommonHistoryLeaves.lean`, `leafFintype` gives finiteness and `leafWeight_positive` gives positivity of the product of branch weights for each leaf. Finally, `Walls/RootProducts.lean` defines `LaurentPositive.NatCoeff` to mean equality with a natural number and `LaurentPositive.Positive` coefficientwise using this predicate. It is therefore integer positivity, not just an order condition over the rationals.

A.4. **Notation and scope of the formalized input.** The total coloring degree N in this paper is the homogeneous degree of an elementary expression. It is not the number ℓ of weighted vertices, nor the triangular size D. The source's triangular base at degree N is our Na_0, and its root-degree condition is membership in the integral monoid (**??**). A leaf's anchor-coordinate profile is our $\mu(\eta)$. The target used in the new proof is $R_{\mathbf{m}}$, whereas the unweighted graph extraction uses $\sum_i u_{b(i)}$.

These identifications delimit the proof dependency. The graph realization and positive common-history statements are imported from the cited source; the multiplicity coefficient identity, its finite support argument, and the Dyck applications are proved in this paper. Only the imported graph-realization and common-history machinery is used as formalized input. The new weighted extension and the Dyck applications are proved here in ordinary mathematical form; no Lean formalization of those extensions is claimed.

Appendix B. Exact computational checks

Two finite enumerations provide independent checks of the formulas, including the distinction between ordered monomials and their symmetric specializations. All arithmetic is integral, except for exact rational arithmetic in the change-of-basis calculation. None of these computations is used in the proofs.

B.1. **Weighted natural unit interval graphs.** For a fixed G, weight vector $\mathbf{m}$, and composition $\alpha = (\alpha_1, \dots, \alpha_r)$ of N, the coefficient $[\mathbf{x}^\alpha]C^-_{G,\mathbf{m}}(\mathbf{x}; u)$ can be computed by processing the colors in increasing order. Let $P_a(\mathbf{b})$ record the partial colorings using colors $1, \dots, a$, with vertex i having received b_i colors. Here $0 \le b_i \le m_i$, and $P_0(\mathbf{b})$ is 1 for $\mathbf{b} = \mathbf{0}$ and 0 otherwise. For an independent set I, write $\mathbf{1}_I$ for its incidence vector and put

$$\Delta_G(\mathbf{b}, I) = \sum_{\substack{i<j \\ ij \in E(G)}} \mathbf{1}_{\{i \in I\}} b_j.$$

The recurrence is

$$P_a(\mathbf{b}) = \sum_{\substack{I \subseteq [\ell] \text{ independent} \\ |I| = \alpha_a,\ \mathbf{1}_I \le \mathbf{b}}} u^{\Delta_G(\mathbf{b}-\mathbf{1}_I, I)} P_{a-1}(\mathbf{b} - \mathbf{1}_I). \tag{B.1}$$

The exponent counts precisely the new inversions between color a at a lower-index vertex and an earlier color at an adjacent higher-index vertex. Thus the desired coefficient is $P_r(\mathbf{m})$, without any assumption of symmetry.

We enumerated all weakly increasing extensive functions h on $[\ell]$ and all positive weight compositions $(m_1, \dots, m_\ell)$ with total weight at most six. The number of weighted cases at each total weight is

Total weight N	1	2	3	4	5	6	Total
Weighted graphs	1	3	10	36	137	543	730

For each degree, the elementary-to-monomial change-of-basis matrix was computed by counting zero-one matrices. Exact inversion then gave candidate elementary coefficients. These candidates were checked against every positive composition of that degree, not only against the partition representatives used to solve the system. Zero exponents merely skip colors and do not affect their comparisons. The tests also checked coefficientwise nonnegativity, palindromicity with the exact value of A, and the clique blowup identity.

The enumeration verified 19,903 ordered monomial expansions and 6,857 nonzero partition–statistic coefficient positions across the 730 cases. No discrepancy was found.

B.2. **Direct step-τ factorizations.** A separate enumeration considered multisets with entries in $\{0, 1, 2, 3\}$, total size at most five, and $\tau \in \{0, 1, 2, 3\}$. For each distinct multiset word, the pair statistic (**??**) was evaluated directly and compared with the static decomposition. All weakly increasing factor labels in four color variables were then enumerated. The affine and dual inequalities were checked separately, and the resulting ordered monomials were weighted by the directly computed statistic.

These sums were compared with the predicted elementary and homogeneous expansions. The homogeneous products were expanded by enumerating weak compositions for each factor h_k; the affine comparison therefore did not assume the involution identity. This computation checked 500 multiset–step cases, 5,456 word statistics, and 271,040 candidate word–factor-label combinations. The dual and affine comparisons comprised 39,016 ordered monomial polynomial checks. No discrepancy was found, including in the repeated-value grading shifts.

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